*Activity #2 – Operators*

**Lanze Matthew C. Jallorina** **BSIT 2-1**

**PART I:**

1. Create a Java program using bitwise operators that will produce the output: true, 10, false, 15.

Code:

```java
public class BitwiseOperators {
    public static void main(String[] args) {
        int a = 5;      // binary: 0101
        int b = 10;     // binary: 01010
        int c = 13;     // binary: 01101

        // Bitwise AND (&)
        System.out.println((a & c) > 0); // 0101  & 01101 = 0101 (5 > 0) = true.

        // Bitwise XOR (^)
        System.out.println(b ^ 0);       // 1010 ^ 0000 = 1010 = 10.

        // Bitwise NOT (~)
        System.out.println((~a) == a);   // ~0101 = -6, -6 == 5 = false.

        // XOR Assignment (^=)
        a ^= b;
        System.out.println(a);          // 0101 ^ 1010 = 1111 = 15

    }
}
```

Output:

```
Run  BitwiseOperators

"C:\Program Files\Java\jdk-25\bin\java.exe" "-j
true
10
false
15

Process finished with exit code 0
```

2. Create a Java program using the ternary operator that will produce the output: true, false, 25, 30.

Code:

TernaryOperators.java

```java
public class TernaryOperators {
    public static void main(String[] args){
        int a = 25;
        int b = 30;

        // true
        System.out.println(a < b ? true : false);

        // false
        System.out.println(a > b ? true : false);

        // 25
        System.out.println(a < b ? a : b);

        // 30
        System.out.println(a > b ? a : b);
    }

}
```

Output:

```
"C:\Program Files\Java\jdk-25\bin\java.exe"
true
false
25
30

Process finished with exit code 0
```

3. Create a Java program using assignment operators that will produce the output: Addition: 120, Subtraction: 70, Multiplication: 144, Division: 16.

Code:

AssignmentOperators.java

```java
public class AssignmentOperators {
    public static void main(String[] args) {
        int b = 24;
        int x;

        // Addition Assignment (+=)
        x = 96;
        x += b;
        System.out.println("Addition: " + x);

        // Subtraction Assignment
        x = 94;
        x -= b;
        System.out.println("Subtraction: " + x);

        // Multiplication Assignment
        x = 6;
        x *= b;
        System.out.println("Multiplication: " + x);

        // Division Assignment
        x = 384;
        x /= b;
        System.out.println("Division: " + x);
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-25\bin\java.exe"
Addition: 120
Subtraction: 70
Multiplication: 144
Division: 16

Process finished with exit code 0
```

**Part II:**

Code:

```java
public class Operations {
    public static void main(String[] args){
        int p = 7;
        int q = 3;
        int r = 15;
        int s = 8;
        int t = 20;

        // Output 1: Unary (++)         Outputs 16
        System.out.println(p++ + ++p);

        // Output 2: Assignment (*=)    Outputs 45
        q *= r;
        System.out.println(q);

        // Output 3: Relational (>)     Outputs true
        System.out.println(q > t);

        // Output 4: Logical (!)        Outputs true
        System.out.println(!(s == q));

        // Output 5: Bitwise (^)        Outputs 7
        System.out.println(r ^ s);

        // Output 6: Unary (--)         Outputs 38
        System.out.println(--t + t++);
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-25\bin\java.exe"
16
45
true
true
7
38

Process finished with exit code 0
```